

UNIT $\rightarrow$ 2

Date

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$\rightarrow$ Characteristic Equation

Given a square matrix A of order ' n '. The characteristic eq is obtained from

$$\det(A - \lambda I) = 0$$

This is a polynomial eq of degree n in ' λ '

Q. Find characteristic polynomial & eq for the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$(2-\lambda)^2 - 1 = 0$$

$$4 + \lambda^2 - 4\lambda - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

Char Poly $\Rightarrow \lambda^2 - 4\lambda + 3$

Char eq $\rightarrow \lambda^2 - 4\lambda + 3 = 0$

$\rightarrow$ Eigen Value

An eigen value of a matrix ' A ' is a scalar λ that satisfies the characteristic eqⁿ or eigen value is the root of characteristic eqⁿ

Q Find eigen value of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$\lambda^2 - 3\lambda - \lambda + 3 = 0$$

$$\lambda(\lambda - 3) - 1(\lambda - 3) = 0$$

$$\lambda = 1 \quad \& \quad \lambda = 3$$

1, 3 $\rightarrow$ Eigen values

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Q C-eg. λ eigen-value for $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} 1-\lambda & 0 & 1 \\ 2 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

$$1-\lambda (1+\lambda^2-2\lambda) + 1(0)$$

$$1+\lambda^2-2\lambda-1-\lambda^3+2\lambda^2=0$$

$$1+3\lambda^2-3\lambda-\lambda^3=0$$

$$\lambda^3+3\lambda-3\lambda^2-1=0$$

$$(\lambda-1)^3=0$$

$$\lambda-1=1$$

→ Eigen Vector

Corresponding to each eigen value '1', the eigen vector is a non-zero vector v such that $(A-\lambda I)v=0$

Q Eigen vector to each λ for $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$

$$\begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 3 \\ 0 & 0 & 4-\lambda \end{bmatrix} \quad \begin{aligned} (1-\lambda)(2-\lambda)(4-\lambda) - 2(0) \\ 3(1-\lambda)0(2-\lambda)(4-\lambda) = 0 \\ 0+1=1 \quad \lambda=2 \quad \lambda=4 \end{aligned}$$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2b \\ b+3c \\ 3c \end{bmatrix} = 0$$

$$v_1 = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -a+2b \\ 3c \\ 2c \end{bmatrix} \quad a=2b$$

$$\Rightarrow v_2 = \begin{bmatrix} 2b \\ b \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 2 & 0 \\ 0 & -2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -3a+2b \\ -2b+3c \\ 0 \end{bmatrix}$$

$$b = \frac{3a}{2}$$

$$3c = 2b$$

$$3c = 3a$$

$$a = c$$

$$\Rightarrow v_3 = \begin{bmatrix} a \\ 3a/2 \\ a \end{bmatrix}$$

→ Eigen space

collection of the possible eigen vectors corresp to eigen value 1 & denoted by E_1

$$E_{\lambda=1} \rightarrow \left\{ (t, 0, 0) : t \in \mathbb{R} \atop t \neq 0 \right\}$$

$$E_{\lambda=2} \rightarrow \left\{ (2t, t, 0) : t \in \mathbb{R} - \{0\} \right\}$$

$$E_{\lambda=4} \rightarrow \left\{ (t, \frac{3t}{2}, t) : t \in \mathbb{R} - \{0\} \right\}$$

→ Cayley Hamilton Theorem

Every square matrix satisfies its own characteristic equation

Q Verify Cayley Hamilton Theorem for the matrix

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \text{ \& Find } A^{-1}$$

$$\begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix} \quad (1-\lambda)(3-\lambda) - 8 = 0$$

$$1^2 - 4\lambda - 5 = 0$$

Put $\lambda = A$

LHS

$$A^2 - 4A - 5I$$

$$A^2 = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix}$$

$$\begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$RHS = \begin{bmatrix} I_2 - 0 \\ I_2 - 0 \end{bmatrix} = \begin{bmatrix} I_2 \\ I_2 \end{bmatrix}$$

$$LHS = RHS$$

Hence Verified

$$A^2 - 4A - 5I = 0$$

$$5I = A^2 - 4A$$

$$5A^{-1} = A - 4I$$

$$5A^{-1} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

8 Express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$

reduce to lowest forms

$$A^2 = 4A + 5I$$

$$A^4 = 16A^2 + 25I + 40A \quad \text{① Divide by char}$$

$$A^4 = 16(4A + 5I) + 25I + 40A \quad \text{eqn 1}$$

$$A^4 = 64A + 80I + 25I + 40A \quad \text{② Remainder this}$$

$$A^4 = 104A + 105I$$

$$A^5 = 104A^2 + 105A$$

$$A^5 = 104A(4A + 5I) + 105A$$

$$A^2 - 4A - 5I = 0$$

$$\sqrt{\begin{array}{r} A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I \\ A^5 - 4A^4 - 5A^3 \hline \end{array}} \quad A^3 - 2A + 3$$

$$A^5 - 4A^4 - 5A^3$$

$$-2A^3 + 11A^2 - A - 10I$$

$$-2A^3 + 8A^2 + 10A$$

$$3A^2 - 11A - 10I$$

$$3A^2 - 12A - 15I$$

$$A + 5I$$

$$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I = \underbrace{(A^2 - 4A - 5I)}_0 (A^3 - 2A + 3) + \underline{A + 5I}$$

$$\Rightarrow A + 5I$$

$$\underline{A + 5I}$$

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→ Properties of eigen values

Suppose λ is the eigen value of a sq matrix A

- $\frac{1}{\lambda}$ is the eigen value of A^{-1}
- λ^n is the eigen value of A^n
- If λ is eigen value of B then $(1+\lambda)$ is eigen value of $(A+B)$ [if A & B commute & simultaneous diagonalisable]
ie. $AB = BA$
- $\frac{|A|}{\lambda}$ is eigen value of $\text{Adj}(A)$
- A & A^T have same eigen values & characteristic polynomial

$$\underline{Q} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 3 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A \begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 9-\lambda \end{bmatrix} \quad B \begin{bmatrix} 3-\lambda & 3 & 1 \\ 1 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

$$(1-\lambda)^2(9-\lambda) = 0$$

$$\lambda = 1, 9$$

$$(3-\lambda)(1-\lambda)^2 - 3(1-\lambda) = 0$$

$$(1-\lambda)[(3-\lambda)(1-\lambda) - 3] = 0$$

$$(1-\lambda)[3 - 4\lambda + \lambda^2 - 3] = 0$$

$$(1-\lambda) \cdot \lambda(1-4) = 0$$

$$\lambda = 0, \lambda = 1, \lambda = 4$$

$$A+B = \begin{bmatrix} 4 & 5 & 4 \\ 1 & 2 & 4 \\ 0 & 0 & 10 \end{bmatrix} \quad \begin{bmatrix} 4-\lambda & 5 & 4 \\ 1 & 2-\lambda & 4 \\ 0 & 0 & 10-\lambda \end{bmatrix}$$

$$(4-1)(2-1)(10-1) - 5(10-1)$$

$$(10-1) [8 - 6 + 1^2 - 5]$$

$$(10-1) (1^2 - 8 + 3)$$

$$\lambda = 10 \quad \Delta \lambda = 3 \pm \sqrt{6}$$

Property 3 nahi hui satisfy $\therefore AB \neq BA$

Also not diagonalised

→ Property

→ Sum of all eigen value = trace of matrix

→ Product of all eigen values = determinant of the matrix

→ Diagonalisation

$$A = PDP^{-1}$$

A & D → similar matrix

A sq matrix 'A' is said to be diagonalisable if it is similar to a diagonal matrix D i.e. A can be written as PDP^{-1} where

P is invertible matrix whose columns are eigen vectors of A & D is diagonal matrix whose diagonal entries are eigen values of A

→ Similar matrices

→ If A & B are 2 similar matrices then

$$\text{Rank}(A) = \text{Rank}(P(A)) = \text{Rank}(B)$$

$$\text{Tr}(A) = \text{Tr}(B)$$

$$|A| = |B|$$

$$\text{Eigen values of } A = \text{Eigen values of } B$$

$$\text{Char. eq of } A = \text{Char eq of } B$$

$$\text{Nullity of } A = \text{Nullity of } B$$

Q Check whether A can be diagonalised

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$12 - 7\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 2)(\lambda - 5) = 0$$

$$\lambda = 2, 5$$

If $\lambda = 2$

$$(A - 2I)V_1 = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2a_1 + b_1 \\ 2a_1 + b_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow a_1 = -\frac{b_1}{2}$$

Let $V_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ & $V_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\Rightarrow P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \quad \& \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\Rightarrow P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ +2 & 1 \end{bmatrix}$$

If $\lambda = 5$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2a_2 + 2b_2 \\ 2a_2 - 2b_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow a_2 = b_2$$

$$\Rightarrow PDP^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ +2 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 5 \\ -4 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 12 & 3 \\ 6 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = A$$

$\Rightarrow A$ is diagonalisable

$$A \sim D$$

$$Q \quad A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 0 & -\lambda & 1 \\ 0 & 0 & -\lambda \end{vmatrix} \quad \begin{aligned} -\lambda(\lambda^2) - 1(0) &= 0 \\ \lambda^3 &= 0 \\ \lambda &= 0 \end{aligned}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

$$\begin{bmatrix} b+c \\ c \\ 0 \end{bmatrix} = 0$$

$$c = 0$$

$$b = 0$$

$$a \rightarrow \text{arbitrary}$$

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad V_2 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \quad V_3 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|P| = 0$$

~~P isn't invertible~~

~~$\Rightarrow A$ isn't diagonalisable~~

~~P^{-1} can't be found~~